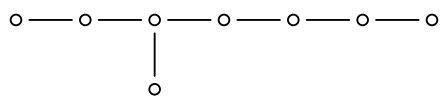

The graphs listed in the theorem are called (simply laced) **Dynkin diagrams**. These graphs arise in a multitude of classification problems in mathematics, such as the classification of simple Lie algebras, singularities, platonic solids, reflection groups, etc. In fact, if we needed to make contact with an alien civilization and show them how sophisticated our civilization is, perhaps showing them Dynkin diagrams would be the best choice!

As a final example consider the representation theory of finite groups, which is one of the most fascinating chapters of representation theory. In this theory, one considers representations of the group algebra $A = \mathbb{C}[G]$ of a finite group G — the algebra with basis $a_g, g \in G$, and multiplication law $a_g a_h = a_{gh}$. We will show that any finite dimensional representation of A is a direct sum of irreducible representations, i.e., the notions of an irreducible and indecomposable representation are the same for A (Maschke's theorem). Another striking result discussed below is the Frobenius divisibility theorem: the dimension of any irreducible representation of A divides the order of G. Finally, we will show how to use the representation theory of finite groups to prove Burnside's theorem: any finite group of order $p^a q^b$, where p, q are primes, is solvable. Note that this theorem does not mention representations, which are used only in its proof; a purely group-theoretical proof of this theorem (not using representations) exists but is much more difficult!

2.2. Algebras

Let us now begin a systematic discussion of representation theory.

Let k be a field. Unless stated otherwise, we will always assume that k is algebraically closed, i.e., any nonconstant polynomial with coefficients in k has a root in k. The main example is the field of complex numbers $\mathbb{C}$, but we will also consider fields of characteristic p, such as the algebraic closure $\overline{\mathbb{F}}_p$ of the finite field $\mathbb{F}_p$ of p elements.